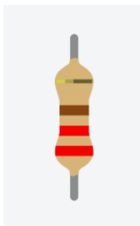
Beat Box Music Machine

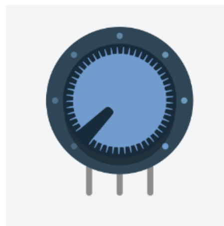
In this project, we'll combine components that use **sound, light,** and **input** together. The circuit will be a **light-reactive beatbox**.

Materials:

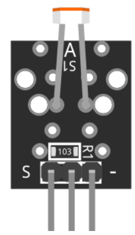
LED:
3 x any color



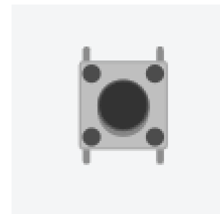
Resistor:
3 x 220Ω



1 x Variable Resistor



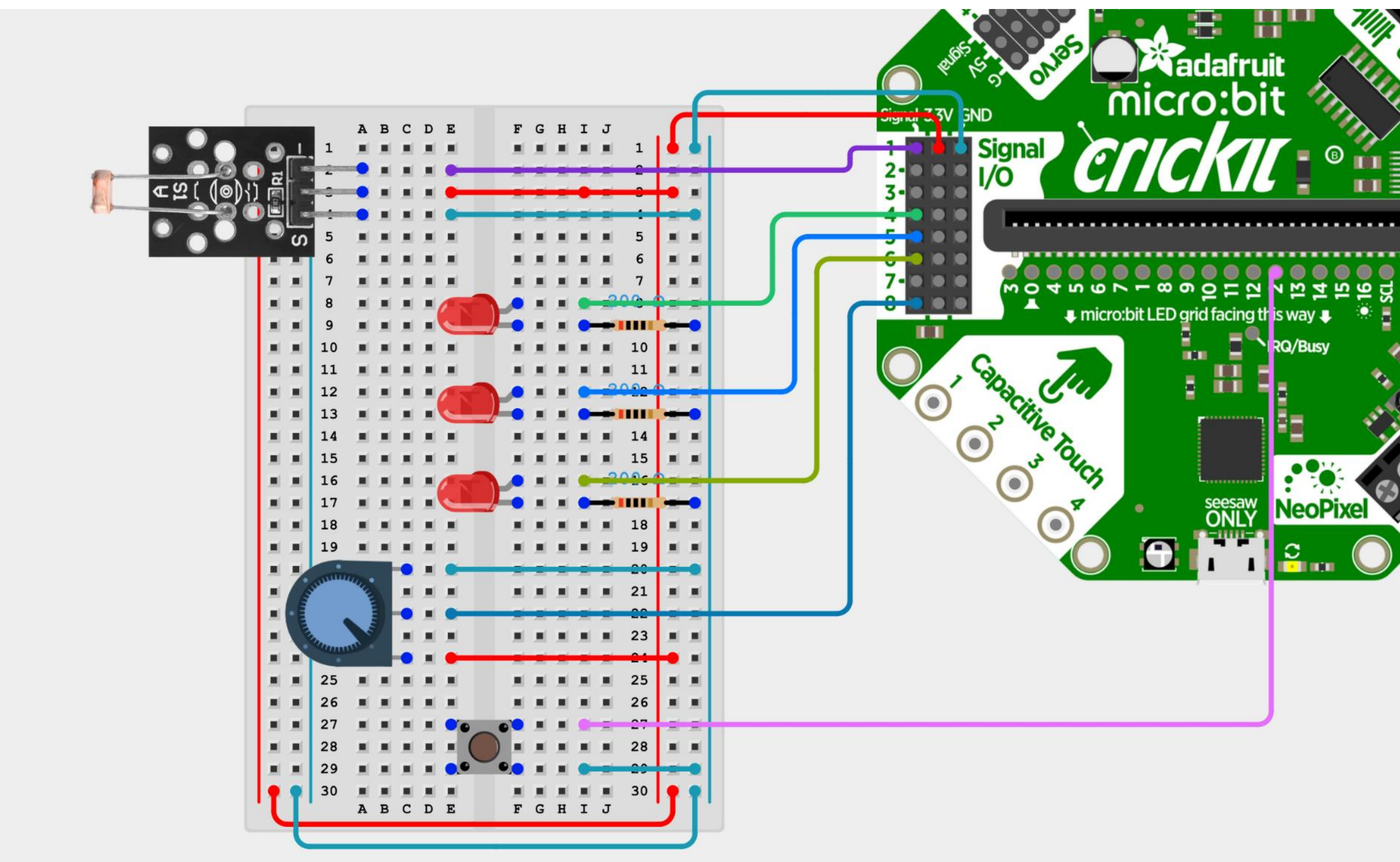
1 x Photoresistor Breakout Board



1 x Pushbutton

Wiring Diagram:

Wire your project according to the following diagram.



Verify Circuit

As the complexity of this circuit is increasing, so is the possibility of errors in component placement (or poor connections for the physical circuit). Before getting into the complexity of the whole project, it would be a good idea to **verify that you've wired the components correctly**, by individually running a simple test on each.

Note: *Once you're done a test, disable that code and start on the next test. You shouldn't need to run all the tests concurrently.*

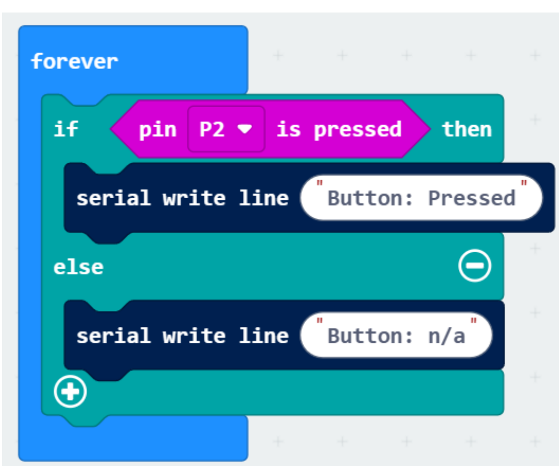
Test One: LEDs

Easy to wire, easy to test. A good place to start.

modify and run this script 3 times (also for signal 5 and signal 6) to ensure the correct LED lights up successfully.

Test Two: Pushbutton

We have one button to test, connected to pin 2. In this circuit you'll want your program to know at all times whether the button is currently pressed or not; simulate that sort of a test with the following script.

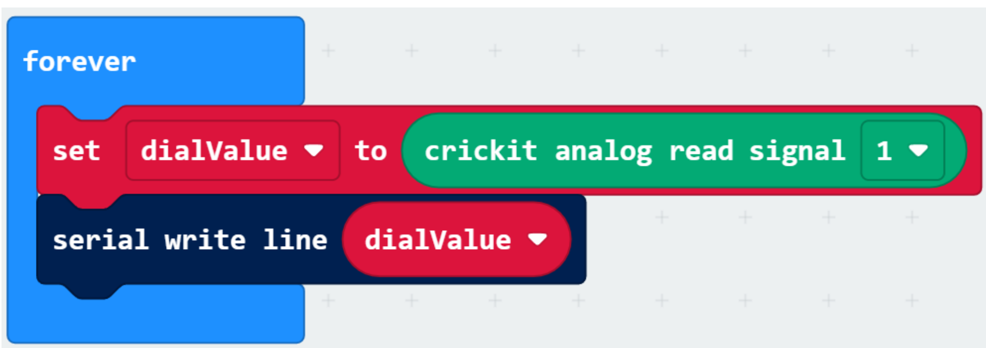


Test Three: Potentiometer

Changing the dial position should give us a different reading. Simplest way to verify is to take a reading and immediately print it to the Serial Monitor:

Test Four: Photoresistor Sensor (KY-018)

Lastly, we should ensure we are able to get valid readings from the light sensor. Try the code below and either shade the sensor with your hand or shine a light on the sensor.

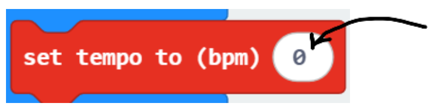


Assignment Details: Base Features

<u>Four on the Floor (Set the beat):</u>

The potentiometer will be used to set the **tempo** (speed) of how fast the music is played. While the program is running, we should continually be updating the value we read from the potentiometer.

There is a block that can be used to set the speed (or tempo) of notes generated:

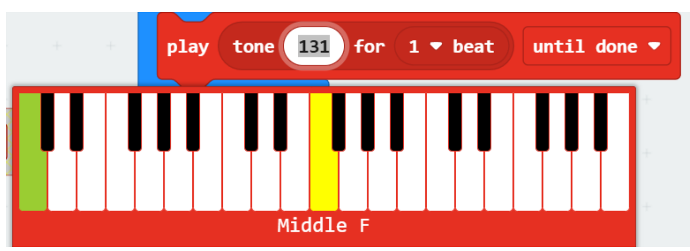


This 0 will be replaced with your *mapped* tempo value

It expects a value between 40 and 500. **Map()** the value of your *tempo* variable and use the mapped value to set the tempo.

<u>Note Selection:</u>

This beat box can play 3 notes. Choose them either by note name or frequency:



Which note is played will depend on the **light sensor value.**

- When testing the photoresistor reading, take note of the range of values you get from the sensor.
- Divide that range up into three even partitions. Each range will be associated with one of your three notes.
- With whichever range is selected, *play that note for 1 beat, then rest for 1 beat.*

Example:

If your sensor readings gave values between 250 and 850, the three even ranges would be:
<u>250-450</u>: lowest note <u>451-650</u>: middle note <u>651-850</u>: highest note

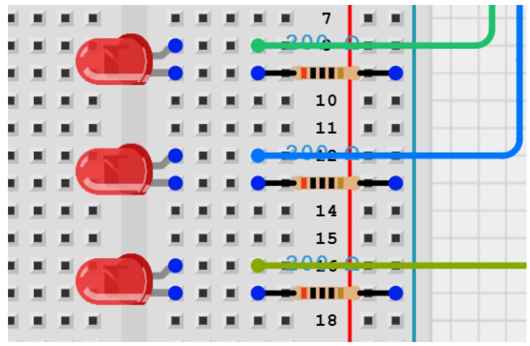
LED Visualization:

Whenever a sound is played, one of the three LED should be lit up as well. During the rests (moments of silence) all three LEDs should be off:

LED 1 → High Note

LED 2 → Middle Note

LED 3 → Low Note



Pitch Shift:

Use of the pushbutton will modify the notes being played according to this scheme:

If the button is not being pressed:

Play music with the regular 3 notes according to the light sensor reading

If the button is being pressed:

Play music with 3 modified (higher) notes according to the light sensor reading.

------ Suggestions and Hints are below but try to build what you can on your own first! ------

Hints and Suggestions: